

A Roadmap of Adaptive Abstraction in Reinforcement Learning, World Models, and Hierarchical Planning

Foundations of Model-Based Reinforcement Learning

The foundational paradigm of Model-Based Reinforcement Learning (MBRL) centers on constructing predictive simulators of environment dynamics to optimize policies or plan action sequences. Standard model-free approaches suffer from high sample complexity, which learned dynamic models mitigate by generating synthetic transitions. However, early formulations struggled with compounding modeling errors, where minor prediction inaccuracies accumulated exponentially over long horizons, leading to catastrophic policy degradation. Resolving this tension has driven key algorithmic advances, emphasizing uncertainty propagation, dynamic model trust bounds, and differentiable planning layers.

Value Iteration Networks

- **Title:** Value Iteration Networks
- **Authors:** Aviv Tamar, Yi Wu, Garrett Thomas, Sergey Levine, Pieter Abbeel
- **Year:** 2016
- **Venue:** Advances in Neural Information Processing Systems (NeurIPS)
- **Summary:** The paper proposes an architecture that embeds a differentiable value iteration calculation inside a convolutional neural network structure, allowing value-based path planning to be optimized end-to-end.
- **Which topic(s) it belongs to:** Coarse-to-Fine Planning, Learned Planners
- **Why it is influential or worth reading:** It proved that structural planning equations can be formulated as neural layers, enabling policy representation to dynamically internalize navigational heuristics.
- **Difficulty:** Beginner

Deep Reinforcement Learning in a Handful of Trials using Probabilistic Dynamics Models

- **Title:** Deep Reinforcement Learning in a Handful of Trials using Probabilistic Dynamics Models (PETS)
- **Authors:** Kurtland Chua, Roberto Calandra, Rowan McAllister, Sergey Levine
- **Year:** 2018
- **Venue:** Advances in Neural Information Processing Systems (NeurIPS)
- **Summary:** This work designs a model-based algorithm combining probabilistic neural network ensembles with trajectory sampling to estimate and propagate epistemic and aleatoric uncertainties during model rollouts.
- **Which topic(s) it belongs to:** Model-Based Reinforcement Learning, Adaptive Prediction Horizons

- **Why it is influential or worth reading:** It provides a blueprint for uncertainty-aware model predictive control, matching the asymptotic execution quality of model-free algorithms with significantly reduced sample budgets.
- **Difficulty:** Intermediate

When to Trust Your Model: Model-Based Policy Optimization

- **Title:** When to Trust Your Model: Model-Based Policy Optimization (MBPO)
- **Authors:** Michael Janner, Justin Fu, Marvin Zhang, Sergey Levine
- **Year:** 2019
- **Venue:** Advances in Neural Information Processing Systems (NeurIPS)
- **Summary:** The authors provide a theoretical analysis of monotonic policy optimization bounds under inaccurate models, proposing a framework that restricts model rollouts to short, branched sequences from real trajectories.
- **Which topic(s) it belongs to:** Model-Based Reinforcement Learning, Variable Rollout Lengths
- **Why it is influential or worth reading:** It established the standard methodology for modern hybrid model-based algorithms, showing that short simulation steps are essential for containing compounding errors.
- **Difficulty:** Advanced

Architecture Name	Core Model Mechanism	Planning / Rollout Horizon	Uncertainty Propagation Method	Primary Limitation
Value Iteration Networks (VIN)	Differentiable CNN recurrence over grid-like lattices	Fixed K iterations of Bellman-style convolutions	None (deterministic planning maps)	Restricted to low-dimensional or grid-world states
PETS	Bootstrap ensemble of probabilistic neural networks	Fixed horizon T evaluated by Random Shooting or CEM	Trajectory Sampling (TS) across ensemble	High computational cost of multi-model propagation
MBPO	Bootstrap ensemble of feedforward dynamics models	Variable/Short branched horizons ($k \leq 15$) from real states	Real-data branching bounds	Relies on a single-step, non-hierarchical dynamics target

Synthesis

This domain seeks to resolve the fundamental trade-off between model-free asymptotic performance and model-based sample efficiency. To prevent policy exploitation from model inaccuracies, research focuses on bounding rollout lengths and characterizing epistemic uncertainty. The common design pattern relies on bootstrap ensembles to estimate prediction variance, using these metrics to scale rollout depths or construct pessimistic value estimates. However, most classical MBRL architectures operate under the strict assumption that dynamics must be represented at a uniform, single-step resolution. The model must predict the exact next step, propagating errors sequentially across the entire planning horizon.

The primary limitation of this pattern is its temporal rigidity; it allocates identical computational budgets to highly predictable, trivial transitions as it does to highly volatile, bifurcation-heavy states. This structural bottleneck leaves open the question of how models can learn to

compress transitions or adaptively skip predictable sequences altogether. In the context of adaptive abstraction, these foundational works show that scaling prediction horizons requires moving away from uniform step-by-step prediction. They demonstrate that uncertainty estimation is a critical trigger for deciding when to truncate, refine, or skip steps during mental simulations.

World Models

World Models frame reinforcement learning as cognitive simulation within a learned predictive model of the environment. Rather than planning in raw observation spaces, these systems learn compact latent spaces that capture the underlying causal structure of transitions. By segregating visual representation learning from the transition dynamics, these models allow planners and policies to run inside highly compact, multi-step simulated imaginations.

Mastering Diverse Control Tasks through World Models

- **Title:** Mastering Diverse Control Tasks through World Models (DreamerV3)
- **Authors:** Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, Timothy P. Lillicrap
- **Year:** 2025
- **Venue:** Nature
- **Summary:** This paper introduces an architecture that encodes high-dimensional images into discrete latent states within a Recurrent State-Space Model, optimizing actor-critic networks inside visual imaginations.
- **Which topic(s) it belongs to:** World Models, Multi-step Latent Prediction
- **Why it is influential or worth reading:** It represents the state-of-the-art in general-purpose world modeling, achieving success across visual, discrete, and continuous action benchmarks using fixed hyperparameters.
- **Difficulty:** Beginner

Joint-Embedding Predictive Architectures for Action-Conditioned Planning

- **Title:** Joint-Embedding Predictive Architectures for Action-Conditioned Planning (EB-JEPA)
- **Authors:** Quentin Assran et al.
- **Year:** 2025
- **Venue:** Facebook Research Technical Report / arXiv
- **Summary:** The framework trains predictive architectures to model dynamics in a learned representation space rather than reconstructive pixel space, preventing representation collapse through regularization constraints.
- **Which topic(s) it belongs to:** World Models, Multi-scale Representations
- **Why it is influential or worth reading:** It formalizes a highly efficient, non-reconstructive world-modeling paradigm that ignores high-frequency, task-irrelevant environmental distractions.
- **Difficulty:** Advanced

Model Name	State Representation	Prediction Target Space	Optimization Objective	Computational Bottleneck
DreamerV3	Discrete Categorical Latents (Categorical RSSM)	Reconstructed Pixel Frames, Rewards, and Terminals	ELBO-style reconstruction and variational loss	High visual decoder overhead during reconstruction
EB-JEPA	Continuous Latent Embeddings	Learned Target Latents (non-reconstructive)	Variance-Invariance-Covariance Regularization (VICReg)	Highly sensitive to collapse; hyperparameter-sensitive

Synthesis

World models are designed to learn rich, compact representations of high-dimensional environments, enabling agents to plan entirely within latent state spaces. The core design pattern uses variational autoencoding or sequence modeling to map raw sensory inputs into stochastic and deterministic state vectors, executing transitions via parallelized recurrent networks. However, traditional architectures assume a uniform temporal granularity. They are constrained to model the environment at a single, fixed tick-rate (typically matched to the environment's control loop) and optimize next-step prediction accuracy. This pixel-level and step-level focus introduces significant computational limitations, as models often dedicate excessive parameter capacity to reconstructing task-irrelevant environmental details. Non-reconstructive approaches like JEPAs mitigate this, yet they still generally assume sequential, single-step transitions in latent space. The primary open challenge in this area is learning multi-scale, hierarchical latents that allow predictions at varying temporal resolutions to coexist. For adaptive abstraction, this means a world model should not only compress observations spatially, but also dynamically structure its latent dynamics to scale its temporal steps, predicting over variable, context-driven horizons.

Hierarchical World Models & Temporal Abstraction

Hierarchical World Models and Temporal Abstraction aim to scale cognitive simulations over long horizons by modeling transitions at multiple timescales. Instead of propagating a single high-frequency sequence, these frameworks structure dynamics into coexisting layers. Slowly updated context, event, or goal latents supervise and organize fast-moving, high-resolution models, allowing the agent to reason about the distant future without compounding errors.

Learning Hierarchical World Models with Adaptive Temporal Abstractions from Discrete Latent Dynamics

- **Title:** Learning Hierarchical World Models with Adaptive Temporal Abstractions from Discrete Latent Dynamics (THICK)
- **Authors:** Christian Gumbsch, Noor Sajid, Georg Martius, Martin V. Butz
- **Year:** 2024
- **Venue:** International Conference on Learning Representations (ICLR)
- **Summary:** This paper proposes THICK, an algorithm that builds a latent model hierarchy by training a lower level to update its context variables sparsely while the higher level

models interactions over context changes.

- **Which topic(s) it belongs to:** Temporal Abstraction, Hierarchical World Models, Multi-scale Representations
- **Why it is influential or worth reading:** It introduces a self-supervised event-predictive mechanism that discovers interpretable temporal boundaries without task-specific structures.
- **Difficulty:** Intermediate

Multi Time Scale State Space

- **Title:** Multi Time Scale State Space (MTS3)
- **Authors:** Vaisakh Shaj, Saleh Gholam Zadeh, Ozan Demir, Luiz Ricardo Douat, Gerhard Neumann
- **Year:** 2025
- **Venue:** International Conference on Learning Representations (ICLR)
- **Summary:** The paper derives a probabilistic framework that propagates states over two coexisting timescales using exact, closed-form inference, allowing detailed predictive control and long-range uncertainty projection.
- **Which topic(s) it belongs to:** Temporal Abstraction, Hierarchical World Models, Multi-scale Representations
- **Why it is influential or worth reading:** It provides a mathematically unified treatment of multi-scale dynamics, bridging sequential recurrent structures with windowed parallelized models.
- **Difficulty:** Advanced

Long Short-Term Imagination

- **Title:** Long Short-Term Imagination (LS-Imagine)
- **Authors:** Anonymous
- **Year:** 2025
- **Venue:** International Conference on Learning Representations (ICLR Submission / Forum)
- **Summary:** This paper proposes LS-Imagine, which extends planning horizons using a long-term model for jumpy state transitions guided by learned affordance thresholds and a short-term model for detailed transitions.
- **Which topic(s) it belongs to:** Variable Rollout Lengths, Hierarchical World Models, Temporal Abstraction
- **Why it is influential or worth reading:** It breaks the paradigm of uniform predictive sequences by showing that agents can dynamically decide to skip highly predictable step sequences using spatial indicators.
- **Difficulty:** Advanced

Algorithm Name	Hierarchy Structure	Abstraction Event Trigger	Level Coordination Method	Latent Dimension Scaling
THICK	Two-level Context RSSM	Change in discrete lower-level context state	Top-down categorical latent conditioning	High-level uses sparse categorical representations
MTS3	Formal Multi-Time	Exact probabilistic	Integrated	Multi-scale

Algorithm Name	Hierarchy Structure	Abstraction Event Trigger	Level Coordination Method	Latent Dimension Scaling
	Scale SSM	window boundaries	closed-form propagation equations	continuous belief states
LS-Imagine	Jumpy dual transition paths	Learned affordance map threshold crossings	Goal-conditioned jumpy transitions	Spatially grounded latent states

Synthesis

Hierarchical world models seek to overcome the temporal horizon limits of flat world models by organizing dynamics into multi-scale temporal pathways. The standard design pattern splits representation into high-frequency state transitions and slowly updated global context models, training the high-level network to predict multi-step macro-transitions. Despite their advantages, most papers still assume static, pre-defined division metrics. They often partition time steps into fixed windows or rely on hard-coded threshold heuristics to define event boundaries.

This introduces significant optimization challenges, as propagating gradients across decoupled timescales can cause severe training instability and representation collapse at the higher levels. A major open question is how to learn fully end-to-end adaptive boundaries that dynamically scale based on task complexity rather than relying on fixed or environment-agnostic steps. This research is directly linked to adaptive prediction, demonstrating how an agent can dynamically contract its representation to navigate chaotic bottlenecks or expand it to skip highly predictable sequences.

Hierarchical RL & Skill Discovery

Hierarchical Reinforcement Learning (HRL) and Skill Discovery scale decision-making to complex, long-horizon tasks by compressing actions into reusable skills. Instead of selecting primitive controls at every step, HRL constructs hierarchies of execution. By training high-level layers to propose abstract targets or select skills, and low-level layers to output actions, HRL simplifies credit assignment and enhances exploration in sparse-reward environments.

Deep Hierarchical Planning from Pixels

- **Title:** Deep Hierarchical Planning from Pixels (Director)
- **Authors:** Danijar Hafner, Kuang-Huei Lee, Ian Fischer, Pieter Abbeel
- **Year:** 2022
- **Venue:** Advances in Neural Information Processing Systems (NeurIPS)
- **Summary:** This research details a two-tier reinforcement learning agent where a high-level manager selects abstract latent target subgoals inside a world model, and a low-level worker outputs primitive actions to meet them.
- **Which topic(s) it belongs to:** Hierarchical Reinforcement Learning, Goal-conditioned Policies
- **Why it is influential or worth reading:** It demonstrates how Hierarchical RL can be cleanly combined with visual world models, showing that latent subgoals can emerge autonomously from training prediction dynamics.

- **Difficulty:** Beginner

Adaptive state-action abstractions via rate-distortion

- **Title:** Adaptive state-action abstractions via rate-distortion
- **Authors:** Fernando E. Rosas
- **Year:** 2026
- **Venue:** arXiv / University of Sussex
- **Summary:** This work proposes soft state-action abstractions developed from rate-distortion principles, refining representation resolution dynamically when the learning error within the abstraction equals the abstraction distortion itself.
- **Which topic(s) it belongs to:** Skill Discovery, Action Sequence Compression, Multi-scale Representations
- **Why it is influential or worth reading:** It introduces a formal information-theoretic certificate to continuously compress states and actions, demonstrating how much information is structurally required to plan optimally.
- **Difficulty:** Advanced

Framework Name	Subgoal / Action Space	Optimization Criteria	Temporal Coordination	Compression Metric
Director	Latent feature vectors within a learned world model	Maximization of task reward and worker intrinsic progress	Static subgoal proposal every H steps	Manual latent spatial bottlenecking
Rate-Distortion Abstractions (ASADI)	Soft stochastic state-action abstract groups	Minimization of Shannon information rates under bisimulation distortion	Continuous, error-driven refinement schedule	Rate-distortion cost function: $I(S; Z_S) + \lambda I(A; Z_A S) + \beta D(d$

Synthesis

This domain seeks to solve the curse of dimensionality in exploration and long-term credit assignment by structuring control into multiple abstraction tiers. The standard design pattern deploys a goal-conditioned architecture where high-level managers set goals, and low-level workers optimize execution based on intrinsic rewards. Most existing methods, however, assume a rigid structural and temporal separation, where the manager operates at a fixed, coarse temporal frequency and the worker runs at the environment's raw control frequency. This rigid setup introduces planning bottlenecks when task demands shift, as the agent cannot dynamically adjust its planning depth or collapse its hierarchy. A key open question is how to enable models to dynamically switch between coarse-grained behavioral execution and fine-grained, step-by-step control. The rate-distortion framework of ASADI offers a compelling theoretical path: it formalizes an error-decomposition certificate, showing that abstraction resolution should adapt based on learning progress, providing a foundation for adaptive action selection.

Latent Actions

Latent Action learning addresses action-space complexity by compressing high-dimensional,

redundant action trajectories into compact representation spaces. In complex robotic domains, the raw control space is highly high-dimensional and non-Markovian. This subfield extracts actionable representations from sequence data, enabling planners to reason over behavioral spaces rather than raw, low-level controls. This approach allows models to learn directly from video datasets that lack explicit control labels.

Learning to Act without Actions

- **Title:** Learning to Act without Actions (Latent Action Policies - LAPO)
- **Authors:** Dominik Schmidt, Mingi Jiang
- **Year:** 2024
- **Venue:** International Conference on Learning Representations (ICLR)
- **Summary:** LAPO leverages a joint inverse dynamics model and a forward dynamics model with a vector-quantized bottleneck to extract latent actions directly from action-free video datasets.
- **Which topic(s) it belongs to:** Latent Actions, Learned Action Representations, Action Sequence Compression
- **Why it is influential or worth reading:** It validated the hypothesis that physical action dynamics can be recovered in a fully unsupervised manner from visual observations, providing a path to utilize internet-scale videos.
- **Difficulty:** Beginner

LAOF: Robust Latent Action Learning with Optical Flow Constraints

- **Title:** LAOF: Robust Latent Action Learning with Optical Flow Constraints
- **Authors:** Xizhou Bu, Jiexi Lyu, Fulei Sun, Ruichen Yang, Zhiqiang Ma, Wei Li
- **Year:** 2026
- **Venue:** IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- **Summary:** This framework integrates a flow decoder that maps latent action representations into dense optical flow vectors, stabilizing optimization in the presence of visual background distractors.
- **Which topic(s) it belongs to:** Latent Actions, Learned Action Representations
- **Why it is influential or worth reading:** It demonstrates how physical motion priors (optical flow) can prevent representation shortcut learning in visual inverse models.
- **Difficulty:** Intermediate

Framework Name	Representation Form	Bottleneck Mechanism	Decoupling Constraint	Execution / Decoding Space
LAPO	Discrete Quantized Vectors (VQ-VAE)	Strict information bottleneck over transitions	Multi-step consistency losses	Fine-tuned policy or supervised mapping
LAOF	Continuous Latents with motion priors	Pseudo-supervised motion flow prediction	Optical flow reconstruction masks	Decoded directly into continuous robotic actions

Synthesis

Latent action architectures are designed to discover highly compact action spaces from sequential observations, bypassing the need for explicit control labels. The common design

pattern relies on a joint autoencoding loop where an inverse dynamics model maps transitions (o_t, o_{t+1}) to a latent space, and a forward dynamics model reconstructs the next state using that latent representation. Despite their utility, most existing models assume single-step state transitions and are highly susceptible to visual shortcut learning. They can compress background distractors instead of true physical dynamics.

This visual susceptibility limits their scalability to open-world scenes, though adding physical motion constraints like optical flow helps isolate the agent's true degrees of control. A key open question is how to compress multi-step, variable-horizon action sequences into adaptive, macro-latent actions. For adaptive abstraction, this means action sequence compression should adapt its temporal duration dynamically. This would allow the model to represent brief high-frequency adjustments as well as extended, multi-second actions within a single, unified latent space.

Coarse-to-Fine & Adaptive Planning

Coarse-to-Fine and Adaptive Planning represents the integration of modeling, abstraction, and control. Instead of searching over fixed horizons or running uniform-depth model rollouts, these planners dynamically adjust their search granularity and predictive depth. By planning coarsely for distant horizons and recursively refining local trajectories, these models minimize inference complexity and mitigate compounding simulator errors.

Adaptive Coarse-to-Fine Subgoal Refinement for Long-Horizon Offline Goal-Conditioned Reinforcement Learning

- **Title:** Adaptive Coarse-to-Fine Subgoal Refinement for Long-Horizon Offline Goal-Conditioned Reinforcement Learning (CFHRL)
- **Authors:** Kaiqiang Ke, Shenghong He, Chengdong Xu, Yuheng Luo, Xiangyuan Lan, Chao Yu
- **Year:** 2026
- **Venue:** arXiv
- **Summary:** This framework dynamically refines distant goals into locally executable targets based on a learned reachability cost metric, stopping recursive substitution when a target is predicted to be directly reachable.
- **Which topic(s) it belongs to:** Coarse-to-Fine Planning, Adaptive Planning Resolution, Goal-conditioned Policies
- **Why it is influential or worth reading:** It introduces a flexible, reachability-bounded planning depth that adjusts target selection on-the-fly rather than relying on a fixed hierarchical scale.
- **Difficulty:** Intermediate

Efficient Robotic Policy Learning via Latent Space Backward Planning

- **Title:** Efficient Robotic Policy Learning via Latent Space Backward Planning (LBP)
- **Authors:** Dongxiu Liu, Haoyi Niu, Zhihao Wang, Jinliang Zheng, Yinan Zheng, Zhonghong Ou, Jianming HU, Jianxiong Li, Xianyuan Zhan
- **Year:** 2025
- **Venue:** International Conference on Machine Learning (ICML)

- **Summary:** LBP trains a latent goal planner that projects task instructions into a distant goal and plans backward toward the current state, ensuring predicted subgoals remain anchored on-task.
- **Which topic(s) it belongs to:** Coarse-to-Fine Planning, Learned Planners
- **Why it is influential or worth reading:** It demonstrates that backward recursive structures suffer from significantly fewer compounding error propagation issues than traditional forward model predictive control.
- **Difficulty:** Intermediate

Intercepting the Future: Latent-Space Predictive World Model for Dynamic VLA Manipulation

- **Title:** Intercepting the Future: Latent-Space Predictive World Model for Dynamic VLA Manipulation (AHEAD)
- **Authors:** Shahram Najam Syed, Arthur Jakobsson, Haoran Hao, Jeffrey Ichnowski
- **Year:** 2026
- **Venue:** arXiv / ICLR Oral Submission
- **Summary:** AHEAD acts as a predict-then-act wrapper for frozen Vision-Language-Action policies, executing a latent feature world model that dynamically halts rollouts when prediction uncertainty exceeds a variance threshold.
- **Which topic(s) it belongs to:** Adaptive Prediction Horizons, Adaptive Computation, Coarse-to-Fine Planning
- **Why it is influential or worth reading:** It presents a concrete implementation of real-time adaptive computation, showing that agents can alter their prediction depth based on the chaotic or deterministic nature of local dynamics.
- **Difficulty:** Advanced

Adaptive Rollout Allocation for Online Reinforcement Learning with Verifiable Rewards

- **Title:** Adaptive Rollout Allocation for Online Reinforcement Learning with Verifiable Rewards (VIP)
- **Authors:** Hieu Trung Nguyen, Bao Nguyen, Wenao Ma, Yuzhi Zhao, Ruifeng She, Viet Anh Nguyen
- **Year:** 2026
- **Venue:** International Conference on Learning Representations (ICLR)
- **Summary:** VIP employs a Gaussian Process model to evaluate the expected gradient variance contribution of task prompts, dynamically distributing rollout budgets to prioritize informative states.
- **Which topic(s) it belongs to:** Adaptive Planning Resolution, Adaptive Computation, Variable Rollout Lengths
- **Why it is influential or worth reading:** It provides a mathematically rigorous formulation of dynamic computational budgeting, routing resources exclusively to prompts with optimal training information.
- **Difficulty:** Advanced

Planner Name	Search Direction	Temporal Refinement Rule	Execution Trigger / Horizon Halt	Computational Overhead
CFHRL	Forward-recursive refinement	Progressively decomposes targets into closer subgoals	Stops when reachability cost is below execution threshold: $D_{\phi}(s, g) \leq \epsilon_{\text{exec}}$	Computes online neural reachability queries recursively
LBP	Backward-recursive projection	Plans subgoals backward from the final target to the current state	Re-plans dynamically at each environmental step	Requires continuous latent backward projections
AHEAD	Forward-linear extrapolation	Extrapolates patch tokens in latent feature space	Truncates rollout when predicted token uncertainty crosses threshold	Minimal; runs as an add-on wrapper over frozen models
VIP	Batched allocation scheduling	Evaluates expected gradient variance dynamically	Distributes computational budgets using GP-guided optimization	Gaussian Process updating and convex optimization

Synthesis

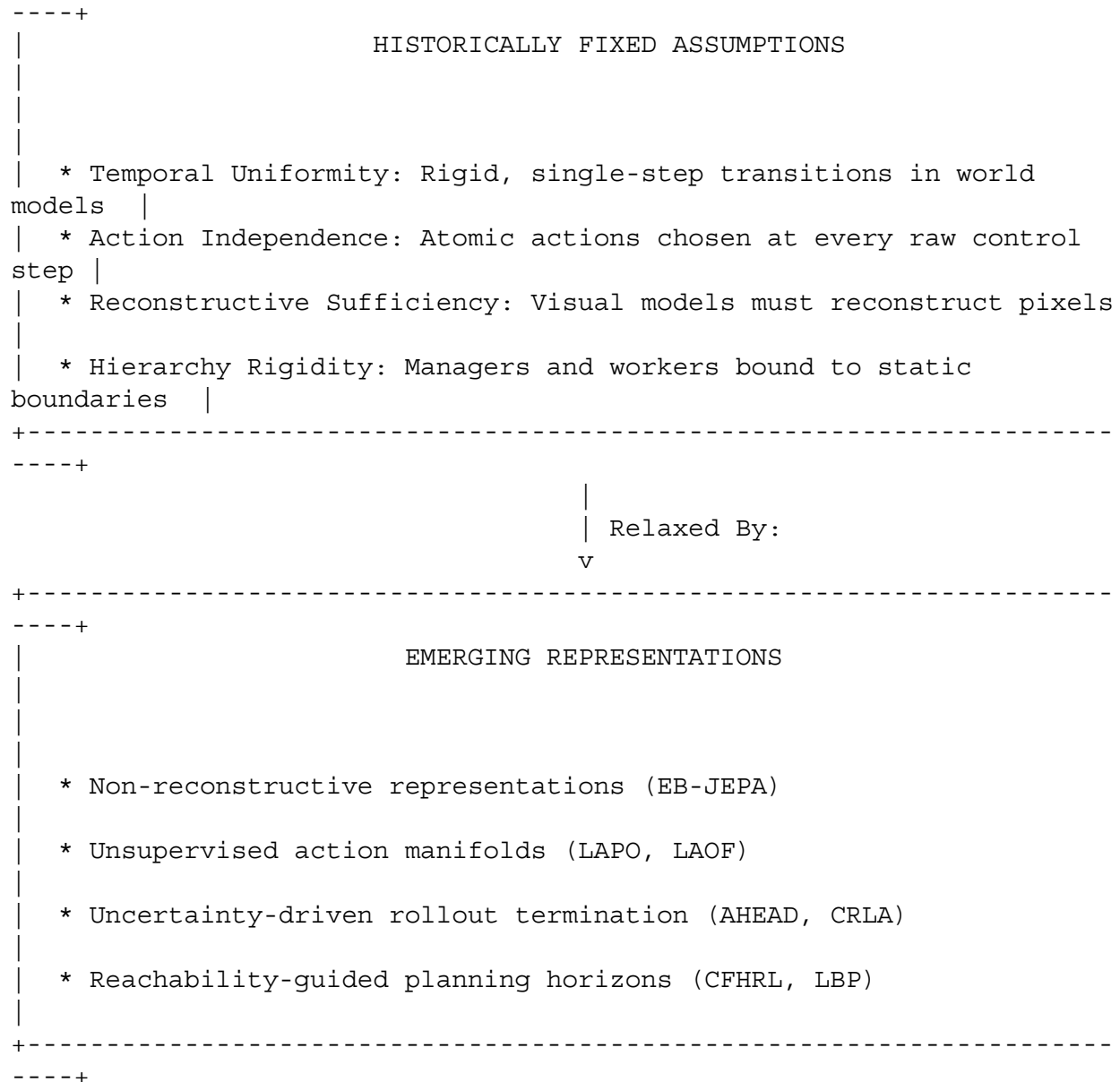
Adaptive planners are designed to solve long-horizon sequential decision problems efficiently by concentrating search effort where it is most needed. The common design pattern uses recursive target decomposition, backward search projections, or uncertainty-gated rollout termination to prioritize search updates. Despite these advances, classical planning pipelines typically assume a fixed directionality and depth of search, applying uniform computation across all states. This rigid approach introduces substantial computational inefficiencies. Planners often allocate identical computational budgets to simple, straight-line trajectories as they do to highly interactive multi-agent scenes. A key open question is how to integrate these adaptive planning strategies into learned, multi-scale world models end-to-end. For adaptive abstraction, these papers provide practical models. They show that planning should adaptively switch between coarse global trajectories and fine-grained local corrections, utilizing estimated reachability, model uncertainty, or gradient variance as triggers for refinement.

Final Synthesis

An analysis of the combined literature reveals several deeply entrenched assumptions that have historically structured reinforcement learning, world models, and hierarchical planning. While many of these assumptions are starting to be relaxed, they highlight the boundaries of the field and point directly to promising research opportunities.

Recurring Assumptions Shared Across Fields

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Historically, temporal uniformity has been a dominant assumption. Almost all established world models and planners assume that transitions must be predicted and executed at a single, fixed step resolution matching the environment's raw control loop. Action representation has also been assumed to be highly atomic, treating continuous controls as independent variables selected at every discrete tick rather than as continuous, compressed macro-actions. Additionally, early visual world models operated under the assumption of reconstructive sufficiency, requiring the reconstruction of full-pixel frames to learn valid dynamics models. Finally, hierarchical architectures have assumed static hierarchies, forcing managers and workers to interact across rigid, hard-coded temporal intervals.

Emerging Relaxation of Assumptions

The literature has begun to systematically dismantle these structural limitations.

Non-reconstructive dynamics have gained prominence through architectures like joint-embedding predictive models, which prove that planning can be performed in learned latent spaces without pixel-level decoding. The assumption of supervised action spaces has been relaxed by self-supervised latent action methods, which demonstrate that true physical actions can be reconstructed directly from visual transition dynamics in unlabeled videos. Furthermore, uniform planning horizons have been challenged by uncertainty-gated termination strategies and reachability-driven recursive refinement. These frameworks prove that planning depth can adapt in real-time based on the predictability and complexity of local transitions.

Comparatively Mature vs. Underexplored Frontiers

An evaluation of the literature helps distinguish mature, highly optimized subfields from underexplored, open research directions:

- **Mature Areas:**
 - Flat, single-step latent world models (such as RSSM-based architectures).
 - Short-horizon, branched model rollouts for policy fine-tuning.
 - Two-tier rigid hierarchical policy structures with fixed-frequency goal proposals.
 - Unsupervised recovery of single-step actions from observation-only transitions.
- **Underexplored Frontiers:**
 - Latent world models that dynamically adjust their transition step-size based on predictability.
 - Multi-scale joint-embedding world models that allow coexisting, non-reconstructive resolutions.
 - End-to-end integration of rate-distortion compression constraints with continuous visual reinforcement learning.
 - Recursive backward-planning loops within learned latent world models.

Suggested Reading Path for Adaptive Abstraction Research

For a researcher investigating how agents can dynamically scale their representation and execution abstractions, the following progressive reading sequence is recommended:

[Phase 1: Horizon Foundations] ----> [Phase 2: Representation Boundaries]

* MBPO	* DreamerV3
* Value Iteration Networks	
* LAPO	

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[Phase 4: Adaptive Planning] <---- [Phase 3: Multi-Scale Dynamics]

* AHEAD	* THICK
* ASADI (Rosas)	* CFHRL
* LBP	* MTS3

1. **Phase 1: Horizon Foundations:** Start with *MBPO* to understand how model error bounds dictate rollout lengths, and *Value Iteration Networks* to study planning as a differentiable neural process.

2. **Phase 2: Representation Boundaries:** Read *DreamerV3* to learn how to construct categorical latent models, and *LAPO* to understand how action spaces can be compressed and recovered from transitions.
3. **Phase 3: Multi-Scale Dynamics:** Study *THICK* to understand how to segment events sparsely, *CFHRL* to analyze how reachability costs coordinate target distances, and *MTS3* to explore exact multi-scale probabilistic inference.
4. **Phase 4: Adaptive Planning:** Read *LBP* to study recursive backward planning, *ASADI* to master the rate-distortion theory of state-action compression, and *AHEAD* to see a practical implementation of real-time, uncertainty-gated adaptive prediction.

Undergraduate Research Proposals

To help an undergraduate researcher identify accessible entry points, three modular, structurally concrete research projects have been designed based on gaps identified in the surveyed literature.

Project 1: Dynamic Latent Rollout Truncation via Local Value Loss Estimators

- **Core Research Question:** Can we dynamically adapt the imagination rollout horizon of a latent world model by tracking the predicted gradient variance of the value function?
- **Mechanistic Approach:** Instead of using a fixed imagination horizon (typically $H=15$ in *Dreamer*), train a lightweight neural network head (inspired by the "conservator" in *CRLA* or the variance estimation in *VIP*) that maps the latent state z_t to the expected temporal gradient variance of the policy update. During latent imagination, halt the rollout dynamically at step τ if the estimated variance or Bellman residual exceeds a calibrated threshold.
- **Tractability:** Extremely high. The investigator can build directly on the open-source *EB-JEPA* or *DreamerV3* repositories. It requires no physical hardware and can be evaluated on standard benchmark suites like Gym or MiniHack.

Project 2: Reachability-Guided Backward Coarse-to-Fine Planning in Offline Datasets

- **Core Research Question:** How does a reachability-guided backward planner compare to a forward-shooting latent planner when utilizing a highly constrained offline replay buffer?
- **Mechanistic Approach:** Implement a goal-conditioned offline planner that utilizes the recursive backward formulation of *LBP* but coordinates target refinement using the reachability check of *CFHRL* ($D_\phi(s, g) \leq \epsilon$). The investigator will measure planning latency and task success rates on standard offline navigation or manipulation datasets (such as *OGBench*).
- **Tractability:** High. This project is entirely offline, removing the need for active online environment collection. It allows the student to directly evaluate mathematical planning concepts by comparing planning speeds and path optimality against standard baselines.

Project 3: Sequence-Compressed Latent Actions via Temporal

Difference Constraints

- **Core Research Question:** Can we learn latent actions that compress variable-length action sequences rather than single-step transitions?
- **Mechanistic Approach:** Modify the LAPO architecture. Instead of inputting consecutive frames (o_t, o_{t+1}) to the inverse dynamics model, input frames at variable intervals ($o_t, o_{t+\tau}$), where τ is dynamically selected based on optical flow thresholds (leveraging the motion saliency concepts in LAOF or AHEAD). Train the model to encode this variable-duration transition into a unified continuous latent action vector, assessing if this sequence-compressed action space accelerates downstream reinforcement learning.
- **Tractability:** Moderate. The student can leverage existing optical flow libraries (like RAFT) to preprocess offline video datasets and modify the open-source LAPO codebase to accept temporal chunks. It provides an intuitive, hands-on exploration of the boundary between action compression and hierarchical option execution.

Works cited

1. Week 5: Model-Based Methods - Deep RL Course, https://deeprlcourse.github.io/course_notes/model-based/ 2. Model-based Policy Optimization with Unsupervised Model Adaptation, https://proceedings.neurips.cc/paper_files/paper/2020/file/1dc3a89d0d440ba31729b0ba74b93a33-Paper.pdf 3. Improving model-based RL with Adaptive Rollout using Uncertainty Estimation - University of Alberta, http://webdocs.cs.ualberta.ca/~vis/asingh1/docs/adaptive_rollout_cmp659.pdf 4. Deep Reinforcement Learning in a Handful of Trials using Probabilistic Dynamics Models - NIPS, <http://papers.neurips.cc/paper/7725-deep-reinforcement-learning-in-a-handful-of-trials-using-probabilistic-dynamics-models.pdf> 5. Deep Reinforcement Learning in a Handful of Trials using Probabilistic Dynamics Models, <https://www.semanticscholar.org/paper/Deep-Reinforcement-Learning-in-a-Handful-of-Trials-Chua-Calandra/56136aa0b2c347cbcf3d50821f310c4253155026> 6. Hierarchical Planning with Latent World Models - arXiv, <https://arxiv.org/html/2604.03208v1> 7. Imagine Within Practice: Conservative Rollout Length Adaptation for Model-Based Reinforcement Learning | OpenReview, <https://openreview.net/forum?id=cYksYKbf6K> 8. When to Trust Your Model: Model-Based Policy Optimization, <http://papers.neurips.cc/paper/9416-when-to-trust-your-model-model-based-policy-optimization.pdf> 9. [1602.02867] Value Iteration Networks - arXiv, <https://arxiv.org/abs/1602.02867> 10. Soft Value Iteration Networks for Planetary Rover Path Planning - OpenReview, <https://openreview.net/forum?id=Sk4m4zWRb> 11. Transfer Value Iteration Networks, <https://ojs.aaai.org/index.php/AAAI/article/view/6022/5878> 12. GitHub - MIRALab-USTC/RLPapers: Must-read papers on Reinforcement Learning (RL), <https://github.com/MIRALab-USTC/RLPapers> 13. When to Trust Your Model: Model-Based Policy Optimization - Semantic Scholar, <https://www.semanticscholar.org/paper/When-to-Trust-Your-Model%3A-Model-Based-Policy-Janer-Fu/9001698e033524864d4d45f051a5ba362d4afd9e> 14. Model-based Policy Optimization under Approximate Bayesian Inference - Proceedings of Machine Learning Research, <https://proceedings.mlr.press/v238/wang24g/wang24g.pdf> 15. A review of learning-based dynamics models for robotic manipulation | Request PDF,

https://www.researchgate.net/publication/395589831_A_review_of_learning-based_dynamics_models_for_robotic_manipulation 16. ARROW: An Adaptive Rollout and Routing Method for Global Weather Forecasting - arXiv, <https://arxiv.org/html/2510.09734v2> 17. ICLR Poster Learning Hierarchical World Models with Adaptive Temporal Abstractions from Discrete Latent Dynamics, <https://iclr.cc/virtual/2024/poster/18558> 18. Intercepting the Future: Latent-Space Predictive World Model for Dynamic VLA Manipulation, <https://arxiv.org/html/2606.02486v1> 19. A Lightweight Library for Energy-Based Joint-Embedding Predictive Architectures - arXiv, <https://arxiv.org/html/2602.03604v3> 20. GitHub - danijar/dreamerv3: Mastering Diverse Domains through World Models, <https://github.com/danijar/dreamerv3> 21. ICML Poster SENSEI: Semantic Exploration Guided by Foundation Models to Learn Versatile World Models - ICML 2026, <https://icml.cc/virtual/2025/poster/44870> 22. 2023-08-11: Paper Summary: "Mastering Diverse Domains through World Models" - WS-DL, <https://ws-dl.blogspot.com/2023/08/2023-08-11-paper-summary-mastering.html> 23. Exploring the limits of hierarchical world models in reinforcement learning - ResearchGate, https://www.researchgate.net/publication/385589476_Exploring_the_limits_of_hierarchical_world_models_in_reinforcement_learning 24. LEARNING HIERARCHICAL WORLD MODELS WITH ADAPTIVE TEMPORAL ABSTRACTIONS FROM DISCRETE LATENT DYNAMICS - ICLR Proceedings, https://proceedings.iclr.cc/paper_files/paper/2024/file/13b45b44e26c353c64cba9529bf4724f-Paper-Conference.pdf 25. Multi Time Scale World Models | OpenReview, <https://openreview.net/forum?id=fY7dShbtmo> 26. Open-World Reinforcement Learning over Long Short-Term Imagination - OpenReview, <https://openreview.net/forum?id=vzltLaEoDa> 27. Exploring the limits of hierarchical world models in reinforcement learning - PMC, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11538428/> 28. Adaptive Coarse-to-Fine Subgoal Refinement for Long-Horizon Offline Goal-Conditioned Reinforcement Learning - arXiv, <https://arxiv.org/html/2605.28127v1> 29. Auto-Generated Curriculum For Reinforcement Learning - OPUS at UTS, <https://opus.lib.uts.edu.au/bitstream/10453/177612/1/thesis.pdf> 30. Danijar Hafner, <https://danijar.com/> 31. Code for paper "Adaptive state-action abstraction via rate-distortion" - GitHub, <https://github.com/ferosas/adaptive-state-action-abstraction> 32. Adaptive state-action abstractions via rate-distortion - arXiv, <https://arxiv.org/pdf/2606.06123> 33. LAOF: Robust Latent Action Learning with Optical Flow Constraints - arXiv, <https://arxiv.org/html/2511.16407v2> 34. From Imagined Futures to Executable Actions: Mixture of Latent Actions for Robot Manipulation - arXiv, <https://arxiv.org/html/2605.12167v1> 35. ICML Poster AdaWorld: Learning Adaptable World Models with Latent Actions, <https://icml.cc/virtual/2025/poster/45343> 36. LEARNING TO ACT WITHOUT ACTIONS - ICLR Proceedings, https://proceedings.iclr.cc/paper_files/paper/2024/file/27985d21f0b751b933d675930aa25022-Paper-Conference.pdf 37. Learning to Act without Actions | OpenReview, <https://openreview.net/forum?id=rvUq3cxpDF¬elId=L2F7FdAjr3> 38. [2312.10812] Learning to Act without Actions - arXiv, <https://arxiv.org/abs/2312.10812> 39. CORIMP: A Correlation-driven Imputation Approach for Offline Reinforcement Learning with Incomplete Action Data - ResearchGate, https://www.researchgate.net/publication/400064693_CORIMP_A_Correlation-driven_Imputation_Approach_for_Offline_Reinforcement_Learning_with_Incomplete_Action_Data 40. LAOF: Robust Latent Action Learning with Optical Flow Constraints - CVF Open Access, https://openaccess.thecvf.com/content/CVPR2026/papers/Bu_LAOF_Robust_Latent_Action_Learning_with_Optical_Flow_Constraints_CVPR_2026_paper.pdf 41. [2511.16407] LAOF: Robust Latent Action Learning with Optical Flow Constraints - arXiv, <https://arxiv.org/abs/2511.16407>

42. Adaptive Rollout Engine - Emergent Mind, <https://www.emergentmind.com/topics/adaptive-rollout-engine> 43. Parallel Coarse-to-Fine Motion Planning via Discrete Flow Matching for Autonomous Driving - CVF Open Access, https://openaccess.thecvf.com/content/CVPR2026/papers/Xu_WAM-Flow_Parallel_Coarse-to-Fine_Motion_Planning_via_Discrete_Flow_Matching_for_CVPR_2026_paper.pdf 44. Daily Papers - Hugging Face, <https://huggingface.co/papers?q=implicit%20sub-goals> 45. [2605.28127] Adaptive Coarse-to-Fine Subgoal Refinement for Long-Horizon Offline Goal-Conditioned Reinforcement Learning - arXiv, <https://arxiv.org/abs/2605.28127> 46. Efficient Robotic Policy Learning via Latent Space Backward Planning - OpenReview, [https://openreview.net/forum?id=DJiouYdH19&referrer=%5Bthe%20profile%20of%20Dongxiu%20Liu%5D\(%2Fprofile%3Fid%3D~Dongxiu_Liu2\)](https://openreview.net/forum?id=DJiouYdH19&referrer=%5Bthe%20profile%20of%20Dongxiu%20Liu%5D(%2Fprofile%3Fid%3D~Dongxiu_Liu2)) 47. Efficient Robotic Policy Learning via Latent Space Backward Planning - arXiv, <https://arxiv.org/html/2505.06861v1> 48. [2606.02486] Intercepting the Future: Latent-Space Predictive World Model for Dynamic VLA Manipulation - arXiv, <https://arxiv.org/abs/2606.02486> 49. VLAW: Iterative Co-Improvement of Vision-Language-Action Policy and World Model, <https://www.semanticscholar.org/paper/VLAW%3A-Iterative-Co-Improvement-of-Policy-and-World-Guo-Lee/affb9be3dedba7952dfad2fb44a3ccdae909c60d> 50. [2602.01601] Adaptive Rollout Allocation for Online Reinforcement Learning with Verifiable Rewards - arXiv, <https://arxiv.org/abs/2602.01601> 51. ADAPTIVE ROLLOUT ALLOCATION FOR ONLINE REINFORCEMENT LEARNING WITH VERIFIABLE REWARDS - OpenReview, <https://openreview.net/pdf?id=Z5sWYACAop> 52. Fixing That Free Lunch: When, Where, and Why Synthetic Data Fails in Model-Based Policy Optimization - arXiv, <https://arxiv.org/html/2510.01457v1>